



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)
FULL SYLLABUS
17-07-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-III) ALL STAR BATCH

PAPER-1

Time : 3 Hours

Maximum Marks : 258

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **36** pages and that all the **24** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

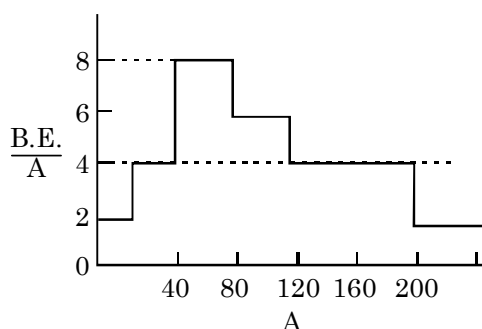
BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I(i) : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

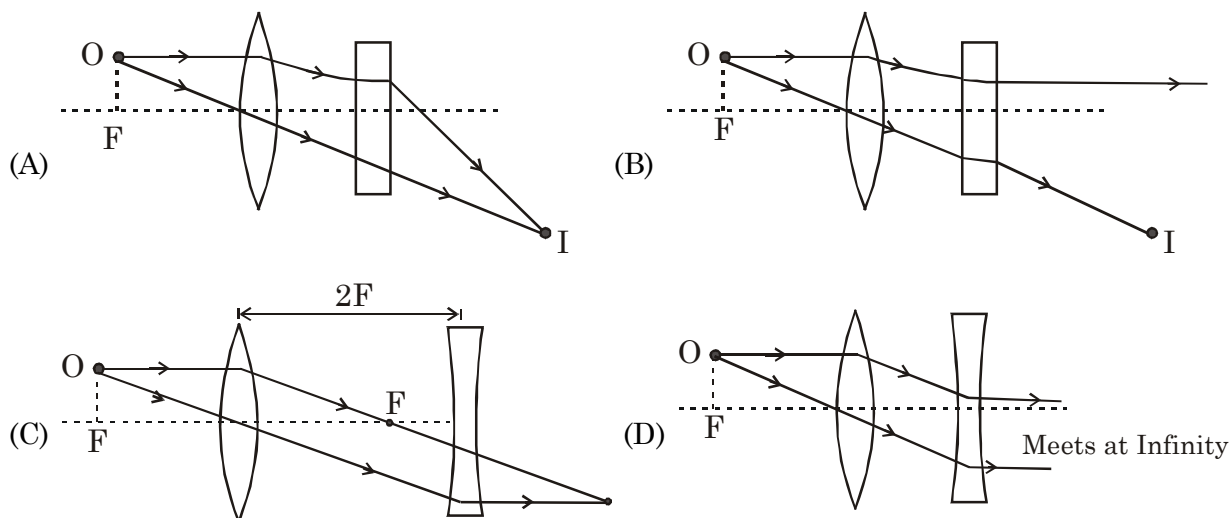
- The following histogram represents the binding energy per particle (B.E./A) in MeV as a function of the mass number A of a nucleus. A nucleus with mass number A = 180 fissions into two nuclei of equal masses. In the process



- (A) 180 MeV of energy is released (B) 180 MeV of energy is absorbed
 (C) 360 MeV of energy is released (D) 360 MeV of energy is absorbed

Space for Rough Work

2. Choose **INCORRECT** ray diagram. All symbols have their usual meaning and all the rays shown are paraxial. (focal length of each lens is F)



3. In presence of a uniform magnetic field $B\hat{j}$ and a uniform electric field $(-E)\hat{k}$, a particle moves undeflected. Which of the following statements is (are) **CORRECT** ?

- (A) The particle can have positive charge, velocity $= -\frac{E}{B}\hat{i}$
 (B) The particle can have positive charge, velocity $= \frac{E}{B}\hat{i}$
 (C) The particle can have negative charge, velocity $= -\frac{E}{B}\hat{i}$
 (D) The particle can have negative charge, velocity $= \frac{E}{B}\hat{i}$

Space for Rough Work

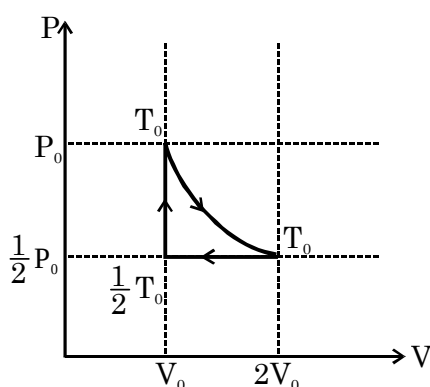
4. The orbiting period of Moon is T_M , orbiting at a height of $(r_M - R_E)$ above the Earth of radius R_E . A geostationary satellite is a satellite that always stays above the same location on the Equator. It stays at a height of $(r_S - R_E)$ above the Earth with an orbiting period of T_S . What is the ratio r_M/r_S in terms of T_M and T_S ?

(A) $(T_M/T_S)^{1/2}$ (B) T_M/T_S (C) T_S/T_M (D) $(T_M/T_S)^{2/3}$

5. One mole of monoatomic ideal gas is initially at pressure P_0 and volume V_0 . The gas then undergoes a three-stage cycle consisting of the following processes:

- (i) An isothermal expansion till it reaches volume $2V_0$, and heat Q flows into the gas
 (ii) An isobaric compression back to the original volume V_0
 (iii) An isochoric increase in pressure till the original pressure P_0 is regained.

The efficiency of this cycle can be expressed as

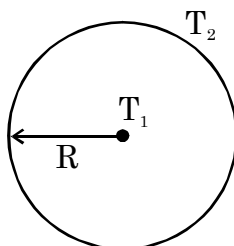


The corresponding P-V diagram is shown above.

(A) $\epsilon = \frac{4Q - 2RT_0}{4Q + 3RT_0}$ (B) $\epsilon = \frac{4Q + 2RT_0}{4Q - 3RT_0}$ (C) $\epsilon = \frac{4Q - 2RT_0}{4Q + RT_0}$ (D) $\epsilon = \frac{4Q + 2RT_0}{4Q + RT_0}$

Space for Rough Work

6. A sphere of radius R is having conductivity K with temperature T_1 at the centre and T_2 at the surface. Assume $T_1 > T_2$. Thermal conductivity K of the sphere varies with radius r as $K = \frac{K_0}{r^2}$. Then select **CORRECT** option(s) for steady state heat flow.



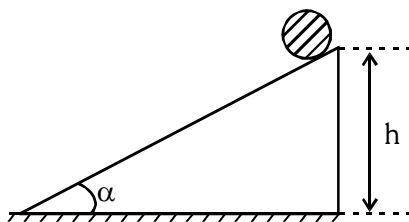
- (A) Heat flows radial outwards.
 (B) The heat current is $\frac{4\pi K_0 (T_1 - T_2)}{R}$.
 (C) Heat can flow in any random manner not necessary in radial direction.
 (D) The heat current is $\frac{2\pi K_0 (T_1 - T_2)}{R}$.
7. The linear mass density of a rod of length L varies from one end to the other as $\lambda_0 \left(1 + \frac{x^2}{L^2}\right)$, where x is the distance from one end, where λ_0 is a constant. The rod is suspended from a ceiling by two massless strings with tensions T_1 and T_2 in them (see figure). Then, which of the following statement(s) is/are **CORRECT** ?



- (A) The mass of the rod is $\frac{2\lambda_0 L}{3}$.
 (B) The center of gravity of the rod is located at $x = \frac{9L}{16}$.
 (C) The tension T_1 in the left string is $\frac{7\lambda_0 Lg}{12}$.
 (D) The tension T_2 in the right string is $\frac{3\lambda_0 Lg}{2}$.

Space for Rough Work

8. A disc of mass 'm' is released from top of inclined plane of slope ' α ' as shown in fig. If ' μ ' is the coefficient of friction between disc and plane surface, choose the **CORRECT** statement(s)



- (A) If $\mu < \frac{1}{3} \tan \alpha$, the acceleration of disc along plane is $(g \sin \alpha)$.
- (B) If $\mu > \frac{1}{3} \tan \alpha$, the net force acting on disc is $\frac{2mg \sin \alpha}{3}$.
- (C) If $\mu > \frac{1}{3} \tan \alpha$, then velocity of disc at the bottom of inclined plane is $\sqrt{\frac{4gh}{3}}$.
- (D) If disc rolls on inclined plane without slipping, then frictional force is $\frac{1}{3} mg \sin \alpha$.

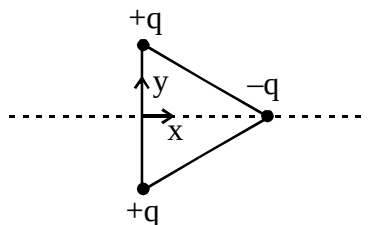
Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 24)

- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **THREE** questions
- Each question has **FOUR** options (A), (B), (C) and (D) **ONE OR MORE THAN ONE** of these four option(s) is(are) correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS.
- Answer to each question will be evaluated according to the following marking scheme:
 - Full Marks* : +4 If only (all) the correct option(s) is (are) chosen.
 - Partial Marks* : +3 If all the four options are correct but **ONLY** three options are chosen.
 - Partial Marks* : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
 - Partial Marks* : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
 - Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered).
 - Negative Marks* : -1 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

Paragraph for Questions 9 to 11

Two positive charges and a negative charge of same magnitude is fixed on vertices of an equilateral triangle. For this set up electric field is investigated on x-axis. Answer following questions for given case.

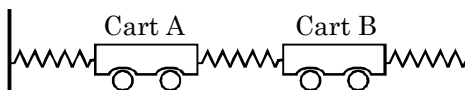


9. Select **CORRECT** statement(s) for electric field and electric potential at origin.
 (A) $E = 0$ (B) $V > 0$ (C) $E \neq 0$ (D) $V < 0$
10. Select **CORRECT** statement(s) for direction of electric field on x-axis.
 (A) Field at origin is towards right.
 (B) Field just right of $-q$ charge is left.
 (C) Field far away on positive x-axis is towards right.
 (D) Field far left on negative x-axis is towards left.
11. Number of points of equilibrium on x-axis is
 (A) 1 (B) 2 (C) 3 (D) 4

Space for Rough Work

Paragraph for Questions 12 to 14

Two identical carts A and B each with mass m are connected via a spring with spring constant k . Two additional springs, identical to the first, connect the carts to two fixed points. The carts are free to oscillate under the effect of the springs in one dimensional frictionless motion. Initially all springs are at relax length.



12. If we remove the middle spring and cart A and B perform SHM (without colliding with each other) then ratio of their time period of A and B is
- (A) 1 (B) $\frac{1}{2}$ (C) 0 (D) ∞
13. Cart A and B (with all three springs) under suitable condition can oscillates in same phase. Lets say they are oscillating in same phase then for this motion select **CORRECT** option(s)
- (A) Length of middle spring remains fixed.
 (B) Length of middle spring varies with time .
 (C) Angular frequency of oscillation of carts is $\sqrt{\frac{k}{m}}$.
 (D) Angular frequency of oscillation of carts is $\sqrt{\frac{3k}{m}}$.
14. Cart A and B (with all three springs) under suitable condition can oscillates out of phase. Lets say they are oscillating out of phase then for this motion select **CORRECT** option(s).
- (A) Length of middle spring remains fixed.
 (B) Length of middle spring varies with time.
 (C) Angular frequency of oscillation of carts is $\sqrt{\frac{k}{m}}$.
 (D) Angular frequency of oscillation of carts is $\sqrt{\frac{3k}{m}}$.

Space for Rough Work

SECTION-II : (Maximum Marks: 30)

- This section contains **TEN** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	●	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	●	●	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

-									
●	●	●	0	0	●	●	●	●	●
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	●	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

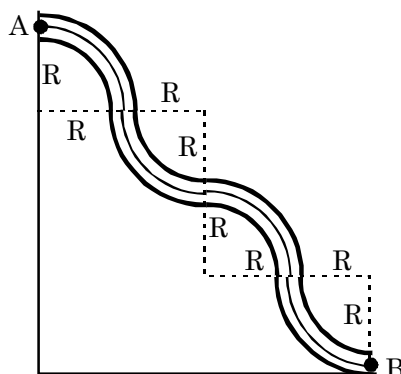
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

- Two travelling waves $3\cos(kx + \omega t)$ and $5\cos(kx - \omega t)$ are travelling in -ve x-axis and +ve x-axis respectively on a string. The magnitude of average energy passing through a point on string per second for resulting wave is of the form $\left(\frac{A_0 \rho \omega^3 S}{k}\right)$. Calculate the value of A_0 . Density of string is ρ and cross-sectional area is S .
- In Young's double slit experiment set-up with light of wavelength $\lambda = 6000\text{\AA}$, distance between the two slits is 2mm and distance between the plane of slits and the screen is 2m. The slits are of equal width. When a sheet of glass of refractive index 1.5 (which permits only a fraction η of the incident light to pass through) and thickness $8000\text{\AA}$ is placed in front of the lower slit, it is observed that the intensity at a point P, 0.15 mm above the initial central maxima does not change. Find the value of η .

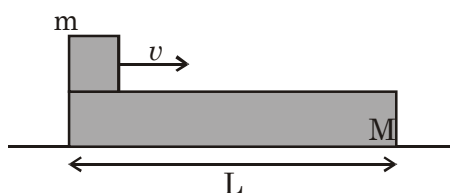
Space for Rough Work

ALLEN

3. A smooth fixed pipe is made-up of four quarter circular segments attached as shown in the figure. A rope of length equal to the length of the pipe is passed through the pipe and is hinged at point A. If the initial acceleration of the rope when released from rest by removing hinge at A is Xg , find the value of $2X$.



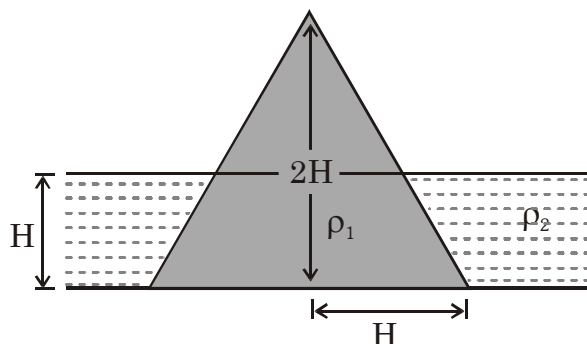
4. As shown in the figure below, a rectangular block of mass M and length L is placed on a smooth table. A small cube of mass m with negligible length is placed on the left end of the block with horizontal initial velocity v sliding to the right. The coefficient of kinetic friction between the block and the cube is μ , and the cube does not fall from the right end of the block. What is the displacement (in m) of the rectangular block when the cube stops sliding on the rectangular block exactly at its right most end ? (Take $m = 2\text{kg}$, $M = 6\text{kg}$ and $L = 10\text{m}$)



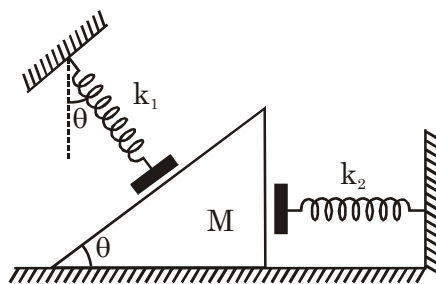
5. A moving hydrogen atom makes a head on collision with another stationary H-atom. Before collision both the atoms are in ground state, and after collision both are getting excited and also move together. The minimum value of the kinetic energy of the incident hydrogen atom is $x(13.6)\text{eV}$. Fill the value of x in OMR sheet.

Space for Rough Work

6. A Liquid ρ_1 is filled inside the cone, liquid ρ_2 is filled outside upto height H as shown in the figure. The light cone is in equilibrium under the action of hydrostatic forces of two liquids of densities ρ_1 and ρ_2 . Find $\frac{\rho_2}{\rho_1}$.



7. A wedge block of mass M is placed on a smooth surface and kept in contact of two unattached springs of force constants k_1 and k_2 . If springs are constrained to deform along length only. Find the time period of small oscillation of wedge on horizontal plane. (given : $M = 4 \text{ kg}$, $k_1 = 16 \pi^2 \text{ N/m}$, $k_2 = \pi^2 \text{ N/m}$ & $\theta = 30^\circ$)



Space for Rough Work

8. A Young's double slit apparatus is immersed in a liquid of refractive index 1.33. It has slit separation of 1 mm and interference pattern is observed on the screen at a distance 1.33 m from plane of slits. The wavelength of light used in air is $6300 \text{ \AA}$. When one of the slit is covered by a glass sheet of thickness $\frac{63}{8n} \mu\text{m}$ and refractive index 1.53, the position of maxima and minima get interchanged. Find the maximum possible value of n .
9. A single electron orbits round a stationary nucleus of charge Ze , where Z is a constant and $-e$ is the electron charge. It is observed that it requires 47.5 eV to excite the electron from second Bohr orbit to third orbit. Then to ionise the atom, (from ground state) required wavelength of electromagnetic (EM) radiations is found to be $n\text{\AA}$. Find the value of n .
10. An inductor $L = 50 \text{ mH}$ carrying an initial current $I_0 = 2.5 \text{ amp}$ is connected across a non linear resistor in which voltage across resistor varies, with current I according to $V = 10 I^2$. How long after (in ms) current through inductor becomes 1.25 amp.



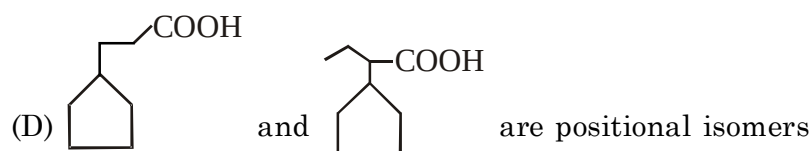
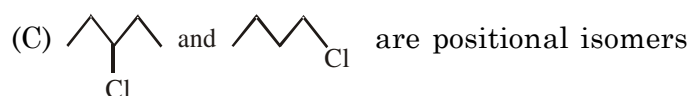
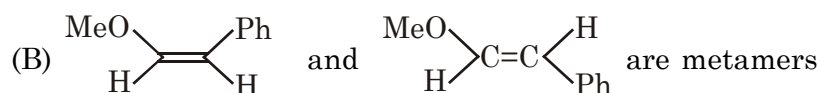
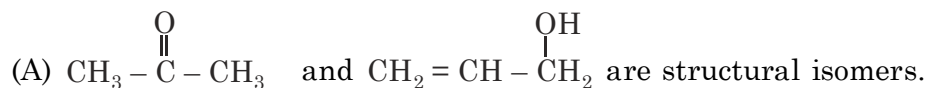
Space for Rough Work

PART-2 : CHEMISTRY

SECTION-I(i) : (Maximum Marks: 32)

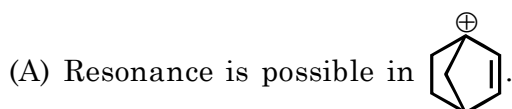
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1. Select **INCORRECT** statement (s) from the following :-



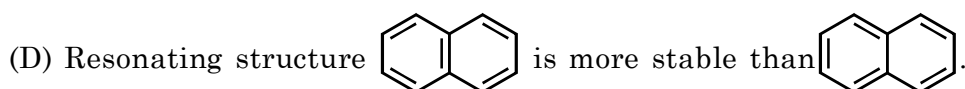
Space for Rough Work

2. Select the **CORRECT** comparison(s) with respect to the indicated properties :
- (A) $O^+ > O > O^-$ (electron affinity)
 (B) $F > O > Se$ (electronegativity)
 (C) $OF_2 > O_2F_2 > H_2O_2$ (electronegativity of central atom)
 (D) $B > N > F$ (electron affinity)
3. The **INCORRECT** statement(s) regarding 2M $MgCl_2$ aqueous solution is/are
 [Given : $d_{\text{solution}} = 1.09 \text{ g/mL}$]
 (A) Molality of Cl^- is 4.44 m
 (B) Mole fraction of $MgCl_2$ is exactly 0.050
 (C) The concentration of $MgCl_2$ is 19% w/v
 (D) The concentration of $MgCl_2$ is exactly $1.9 \times 10^4 \text{ ppm}$
4. Which of the following statement(s) is/are **CORRECT** ?



(B) Resonance is a permanent effect.

(C) Resonance takes place in cyclooctatetraene.



5. Choose the **INCORRECT** option(s) about the oxyacids of non-metallic elements :
- (A) $H_5P_3O_{10}$ has three P – O – P linkages.
 (B) $H_2S_{n+2}O_6$ (polythionic acid) has $(n - 1)$ S – S linkages.
 (C) All $\widehat{OClO}$ bond angles are identical in $HClO_4$.
 (D) HNO_3 is stronger acid than corresponding acid of N_2O_3 .

Space for Rough Work

6. For the reaction.

$$2X + 3Y + 4Z \rightarrow 5W$$
Initially if 1 mole of X, 3 mole of Y and 4 mole of Z is taken. If 1.25 mole of W is obtained under reaction condition and there is no any side reaction, then -
(A) X is limiting reagent (B) % yield of this reaction is 50%
(C) 2 moles of Z is left unreacted (D) 0.75 moles of Y is reacted
7. Which of the following option(s) represent the **CORRECT** IUPAC name of the given co-ordination compounds ?
(A) $\text{Li}[\text{CuMe}_2]$: Lithium dimethylcuprate(II).
(B) $[\text{Os}(\text{Et})(\text{NH}_3)_5]\text{Cl}$: Pentaammineethylosmium(II) chloride.
(C) $\text{K}_3[\text{Co}(\text{CN})_5\text{H}]$: Potassium pentacyanidoprotoniumcobaltate(I)
(D) $[\text{Co}(\text{en})_3]\text{Cl}_3$: Tris-(ethane-1,2-diammine)cobalt(III) chloride.
8. The **CORRECT** option(s) regarding extraction of pure aluminium from red bauxite is/are :
(A) Serpeck's process is used for ore concentration.
(B) Leaching of ore with aq. NaOH produces precipitate of NaAlO_2 .
(C) Weakly acidic medium is used to precipitate $\text{Al}(\text{OH})_3$ from solution containing $\text{Na}[\text{Al}(\text{OH})_4]$.
(D) Only pure Al_2O_3 is used as electrolyte in Hall-Heroult's process of electrolytic reduction.

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 24)

- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **THREE** questions
- Each question has **FOUR** options (A), (B), (C) and (D) **ONE OR MORE THAN ONE** of these four option(s) is(are) correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

Paragraph for Questions 9 to 11

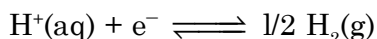
Metal-Insoluble Salt-Anion Half Cell:

In this half cell, a metal coated with its insoluble salt is in contact with a solution containing the anion of the insoluble salt. eg. Silver-Silver Chloride Half Cell:

This half cell is represented as $\text{Cl}^-/\text{AgCl}/\text{Ag}$. The equilibrium reaction that occurs at the electrode is $\text{AgCl(s)} + \text{e}^- \rightleftharpoons \text{Ag(s)} + \text{Cl}^-(\text{aq})$

Gas-Ion Half Cell:

In such a half cell, an inert electrode, platinum or graphite is in contact with gas and a solution containing a specified ion. One of the most important gas-ion half cell is the hydrogen-gas-hydrogen ion half cell. In this half cell, purified H_2 gas at a constant pressure is passed over a platinum electrode which is in contact with an acid solution.



Space for Rough Work

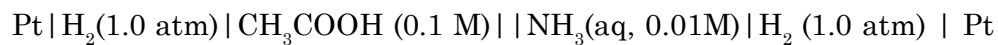
9. E_{cell}^0 for the following cell at 298 K is nearly :



(Given : K_{sp} of AgCl = 2×10^{-10} , K_{sp} of AgBr = 4×10^{-13})

- (A) 0.0 V (B) 0.16 V (C) -0.16 V (D) -0.32 V

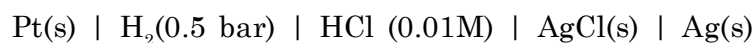
10. Calculate the emf of the cell at 25°C



(Given : $K_{\text{a}}(\text{CH}_3\text{COOH}) = 10^{-5}$, $K_{\text{b}}(\text{NH}_3) = 10^{-5}$)

- (A) -0.44 V (B) -0.88 V (C) -0.22 V (D) -0.11 V

11. For the cell ,



$E_{\text{cell}}^0 = 0.22 \text{ V}$ at 25°C. E_{cell} at 25°C is ($\log 2 = 0.30$)

- (A) 0.0V (B) 0.45 V (C) 0.67 V (D) -0.23 V

Space for Rough Work

Paragraph for Questions 12 to 14

Alkyl halides can undergo S_N1 , S_N2 , $E1$ & $E2$ reactions. Outcome of a reaction is affected by factors like

- | | |
|-------------------------------|------------------------------|
| (1) Nature of solvent | (2) Temperature |
| (3) Structure of alkyl halide | (4) Type of Base/Nucleophile |

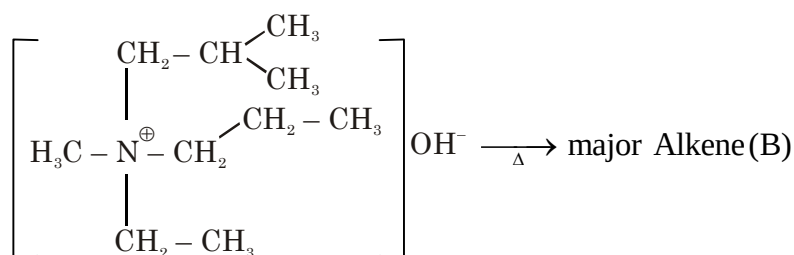
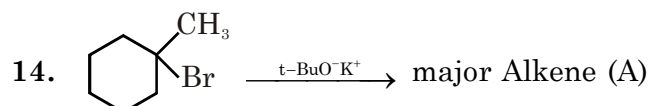
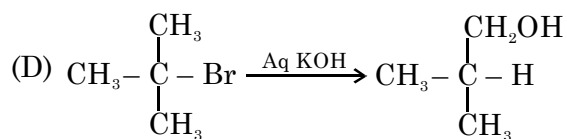
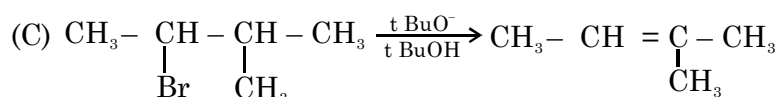
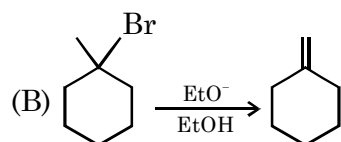
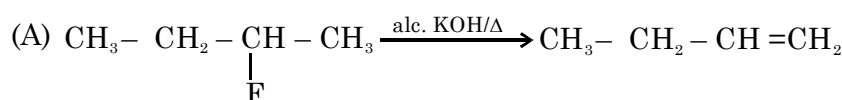
If reaction conditions are altered, outcome of a reaction can change.

Answer following questions.

12. Which of the following compound(s) can give same major product in S_N1 as well as in S_N2 reaction ?

- | | |
|---|---|
| (A) $\text{Ph}-\text{CH}_2-\text{Cl}$ | (B) $\text{CH}_2=\text{CH}-\text{CH}_2-\text{Cl}$ |
| (C) $\text{CH}_3-\underset{\text{CH}_3}{\underset{ }{\text{CH}}}-\underset{\text{Cl}}{\underset{ }{\text{CH}}}-\text{CH}_3$ | (D) $\text{CH}_3-\underset{\text{CH}_3}{\underset{ }{\text{CH}}}-\text{Cl}$ |

13. Which of the following reaction is **CORRECT** for given product as major product ?



Total sum of α -H in Alkene (A) & Alkene (B) is/are

- | | | | |
|-------|-------|--------|-------|
| (A) 4 | (B) 7 | (C) 10 | (D) 8 |
|-------|-------|--------|-------|

Space for Rough Work

SECTION-II : (Maximum Marks: 30)

- This section contains **TEN** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

-									
●	●	●	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

- For reaction $A \rightarrow B$, the rate constant $k_1 = A_1(e^{-E_{a1}/RT})$ and for the reaction $X \rightarrow Y$, the rate constant $k_2 = A_2 e^{-(E_{a2}/RT)}$. If $A_1 = 10^9$, $A_2 = 10^{10}$, $E_{a1} = 1200$ cal/mol and $E_{a2} = 1800$ cal/mol, then the temperature at which $k_1 = k_2$ is $\frac{x}{2.303}$ K. Find $Y = \frac{x}{100}$.
(Given; $R = 2$ cal/K-mol)
- 200 L of a certain liquid is confined in insulated walls at the pressure of 100 atm. The pressure is suddenly released and maintained to 5 atm by which the liquid expanded by 1 % against this external pressure. For the process, the absolute value of $\left(\frac{\Delta U + W}{4}\right)$ in L-atm is.

Space for Rough Work

3. Consider following polymers

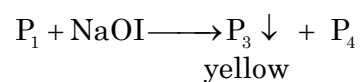
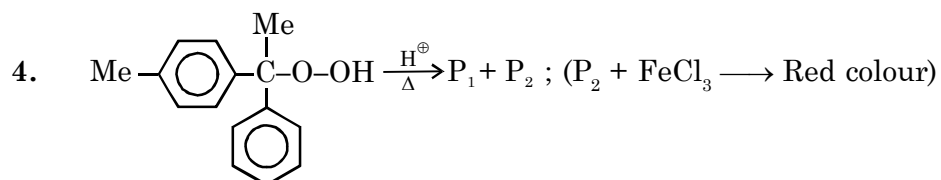
Nylon-6, Nylon-66, Polythene, Polytetrafluoroethene, Buna-S, Cellulose, Cellulose acetate, Cellulose nitrate, Amylose, Glycogen, DNA, RNA

If x = Total number of purely synthetic polymers

y = Total number of semi synthetic polymers

z = Total number of natural polymers

Then $\left(\frac{x^2 + y^2}{z} \right) = \dots\dots\dots$



If x = molecular mass of gas obtained when P_3 is treated with Ag powder.

y = molecular mass of aromatic hydrocarbon obtained when P_4 is treated with sodalime/ Δ .

Then value of $\left(\frac{y}{x} \right)$ is....

Space for Rough Work

5. The number of moles of dihydrogen produced by treating 1 mole of sodium with excess of water = 'X'.
The number of mole of hydrocarbon produced when 1 mole of Mg_2C_3 is reacting with water = Y
The value of (X + Y) would be.
6. What is the total mass of formed soluble manganese compound (in gram) by 3 moles of K_2MnO_4 with aqueous solution having 2 moles of H_2SO_4 ? (atomic weight of Mn = 55, K = 39, S = 32, O = 16, H = 1)
7. Match the list of given species with its ionisation energy and find sum of IE of Helium gas & nitric oxide molecule (in kJ/mole)

(I)**Species****(II)****(Ionisation Energy)**

(A) Helium

(P) 1314 kJ/mole

(B) Nitric oxide molecule

(Q) 1402 kJ/mole

(C) Unbonded atom of nitrogen

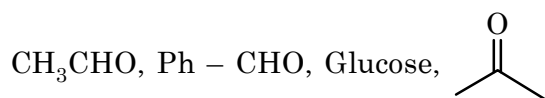
(R) 2372 kJ/mole

(D) Unbonded atom of oxygen

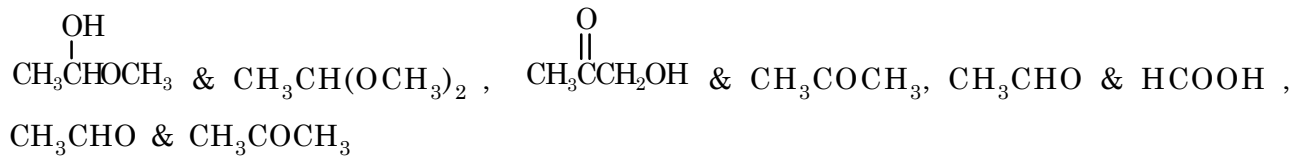
(S) 894 kJ/mole

Space for Rough Work

8. (i) Number of compounds which do not react with Fehling solution are **x** ?



- (ii) Tollen's reagent can be used to distinguish between **y** number of pairs:



[Give your answer as **x + y**]

9. At 298 K the standard Gibbs energy of formation of H₂O(*ℓ*) is –256.5 kJ/mol, while standard Gibbs energy of its ionisation to H⁺ & OH[–] is 80 kJ/mol. What will be emf at 298 K of the cell (volts)?
H₂(g, 1 bar) | H⁺(1M) || OH[–](1M) | O₂ (g, 1 bar)
10. A catalyst lowers the activation energy of a reaction in such a manner that rate constant at 27°C for uncatalysed reaction equals to rate constant at –73°C for catalysed reaction. By how many kJ energy barrier is reduced by catalyst. Activation energy for the uncatalysed reaction is 24 kJ.

Space for Rough Work

PART-3 : MATHEMATICS

SECTION-I(i) : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Consider two circles $S_1 \equiv x^2 + y^2 + 8x = 0$, $S_2 \equiv x^2 + y^2 - 2x = 0$. Let ΔPQR be formed by the common tangents to circles S_1 and S_2 . Then which of the following holds good ?
 (A) Incentre of ΔPQR is (1, 0)
 (B) Equation of Radical axis of S_1 & S_2 is $y = 0$
 (C) Product of slopes of direct common tangents is $\frac{-9}{16}$
 (D) If tranverse common tangent intersect direct common tangents at point Q and R then QR equals 1
2. Which of the following limit(s) vanishes ?

(A) $\lim_{x \rightarrow 0^+} (x^{x^x} - x^x)$

(B) $\lim_{x \rightarrow 0^+} x^2 \ln \sqrt{\frac{1}{x}}$

(C) $\lim_{x \rightarrow 0^+} x^{\ln(x+1)}$

(D) $\lim_{x \rightarrow 0^+} \frac{10^x - 2^x - 5^x + 1^x}{x + \tan x}$

Space for Rough Work

3. If $S_n = \frac{3}{4} + \frac{5}{36} + \frac{7}{144} + \frac{9}{400} + \dots + n$ terms and $P_n = \frac{1}{1 - S_n}$, then
- (A) Unit place digit of P_{40} is 1 (B) Ten's place digit of P_{40} is 8
 (C) Ten's place digit of P_{50} is 0 (D) Ten's place digit of P_{50} is 6
4. On $x - y$ plane, the path defined by equation $\frac{1}{x^m} + \frac{1}{y^m} + \frac{k}{(x+y)^n} = 0$ is a
- (A) parabola if $m = -\frac{1}{2}$, $k = -1$, $n = 0$ (B) hyperbola if $m = 1$, $k = -1$, $n = 0$
 (C) parabola if $m = k = n = 1$ (D) pair of lines if $m = k = -1$, $n = 1$
5. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function such that $f(f(f(x))) = x \forall x \in \mathbb{R}$, then CORRECT option(s) is(are)
- (A) $y = f(x)$ is a bijective function
 (B) $y = f(x)$ is a strictly increasing function
 (C) $y = f(x)$ is strictly decreasing function
 (D) $y = f(x)$ intersects $y = x^2$ at exactly two distinct points.

Space for Rough Work

6. A triangle ABC having vertices A (5, 1), B (−1, − 7) and C (1 , 4) respectively. Let L be the line (mirror) passing through C and parallel to AB. A light ray emanating from point A and goes along the direction of internal bisector of angle A , which meets the mirror at E and BC at D.
- (A) Area of $\triangle CDE = \frac{25}{6}$ (B) Area of $\triangle BCE = \frac{25}{2}$
- (C) Area of quadrilateral ABEC = $\frac{75}{2}$ (D) Area of $\triangle ADC = \frac{25}{9}$
7. If $1 + [\sin^{-1}x] > [\cos^{-1}x]$ (where [] denotes greatest integer function), then x can belong to
- (A) (cos1, sin1) (B) [sin 1,1] (C) [cos 2,0] (D) [cos 1,1]
8. The bisector of $\angle B$ of an isosceles $\triangle ABC$ with $AB = AC$ meets the circumcircle of $\triangle ABC$ at P and meets AC at R and BC produced meet AP at Q. If BC , CA, AB are a , b , c respectively then which of the following is(are) CORRECT ?
- (A) $\triangle CQA$ is isosceles (B) $BR = \frac{2ac}{a+c} \cos B/2$
- (C) $AQ = 2b \cos B/2$ (D) $PQ = BC$

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 24)

- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **THREE** questions
- Each question has **FOUR** options (A), (B), (C) and (D) **ONE OR MORE THAN ONE** of these four option(s) is(are) correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS.
- Answer to each question will be evaluated according to the following marking scheme:
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Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

Space for Rough Work

Paragraph for Questions 9 to 11

A Pythagorean triplet (a, b, c) is such that $a^2 + b^2 = c^2$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 2 \\ 2 & 2 & 3 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 2 & 2 \\ -2 & -1 & -2 \\ 2 & 2 & 3 \end{bmatrix}$$

$$C = \begin{bmatrix} -1 & -2 & -2 \\ 2 & 1 & 2 \\ 2 & 2 & 3 \end{bmatrix}$$

9. If we write pythagorean triplet (a, b, c) in matrix form $[a \ b \ c]$ then which of the following product is a pythagorean triplet ?
 (A) $[3 \ 4 \ 5] A$ (B) $[3 \ 4 \ 5] B$ (C) $[3 \ 4 \ 5] C$ (D) None of these
10. Which of the following(s) hold good ?
 (A) $A^{-1} = \text{adj } A$
 (B) $(AB)^{-1} = \text{adj } (AB)$
 (C) $(BC)^{-1} = \text{adj } (BC)$
 (D) $(ABC)^{-1} \neq \text{adj } (ABC)$
11. $\text{Tr } (A + B^T + 3C)$ equals (Where $\text{Tr } (A)$ denotes trace of square matrix A)
 (A) 17 (B) 15 (C) 19 (D) 18

Space for Rough Work

Paragraph for Questions 12 to 14

A box contains n coins. Let $P(E_i)$ be the probability that exactly i out of n coins are biased. If $P(E_i)$ is directly proportional to $i(i+1)$, $0 \leq i \leq n$

12. Proportionality constant is not equal to

(A) $\frac{3}{n(n^2+1)}$ (B) $\frac{1}{(n^2+1)(n+2)}$ (C) $\frac{3}{n(n+1)(n+2)}$ (D) $\frac{1}{(n+1)(n+2)(n+3)}$

13. What are the odd in favour of a coin selected at random is biased ? (where $n \rightarrow \infty$)

(A) 1 : 3 (B) 3 : 2 (C) 3 : 1 (D) 7 : 1

14. If a coin selected at random is found to be biased then probability that it is the only biased coin in the box

(A) $\frac{1}{(n+1)(n+2)(n+3)(n+4)}$ (B) $\frac{12}{n(n+1)(n+2)(3n+1)}$
 (C) $\frac{24}{n(n+1)(n+2)(2n+1)}$ (D) $\frac{24}{n(n+1)(n+2)(3n+1)}$

Space for Rough Work

SECTION-II : (Maximum Marks: 30)

- This section contains **TEN** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	○	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	②	●	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	⑤	●	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	●	●	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

-									
●	●	●	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	②	●	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	●	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases
-
1. Tangent at P to $xy = c^2$ cuts x-axis & y-axis at M and N respectively. Rectangle OMVN is completed where O is origin. If MV cuts the curve at Q, such that $MV = \lambda MQ$, then λ equals
 2. Let f be a continuous and differentiable function in (x_1, x_2) . If $f(x) \cdot f'(x) \geq x\sqrt{1-(f(x))^4}$ and $\lim_{x \rightarrow x_1^+} (f(x))^2 = 1$ and $\lim_{x \rightarrow x_2^-} (f(x))^2 = \frac{1}{2}$ then minimum value of $x_1^2 - x_2^2$ is λ then $\frac{\lambda}{\pi}$ equals to
 3. The area of the region in Ist quadrant bounded by y-axis, $y = \frac{x}{4}$, $y = 1 + \sqrt{x}$ and $y = \frac{2}{\sqrt{x}}$ is equals to

Space for Rough Work

4. Let $\vec{a} = 2\hat{i} + \hat{j} + \hat{k}$, $\vec{b} = 3\hat{i} + \hat{j} - 5\hat{k}$ and if point of intersection of the lines $\vec{r} \times \vec{a} = \vec{b} \times \vec{a}$ and $\vec{r} \times \vec{b} = \vec{a} \times \vec{b}$ is $\vec{c}$, then $[\vec{a} \vec{b} \vec{c}]$ equals
5. If the line $\frac{x-1}{15} = \frac{y+1}{16} = \frac{z-1}{n}$ intersect the curve $6x^2 + 5y^2 = 20$, $z = 0$ then $9n^2 + 20n - k = 0$, where k equals to
6. The equation $(K - 2)(x - [x])^2 + 2(x - [x]) + K^2 = 0$ (where $K \in \mathbb{R}$) has exactly one solution in $(3, 4)$ (where $[]$ denotes greatest integer function), then $[-K]$ equals

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7. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a function satisfying $|f(x)| \leq 2, \forall x \in [0, 1]$. If $\int_0^1 f(x) dx = 0$ and the greatest value of $\int_0^1 (f(x))^3 dx$ is equal to n , then the value of $\frac{1}{n}$ is
8. Let a, b and c be three real numbers satisfying $1 + a^2 + b^2 + c^2 + (ab)^2 + (bc)^2 + (ca)^2 + (abc)^2 = 25$. If the least value of $(ab + bc + ca)$ is λ , then the value of $|\lambda|$ is

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9. Let a_n denote the number of all n -digit numbers formed by the digit 3, 5 or both such that it
- (i) begins with 3
 - (ii) ends with 3
 - (iii) contains no two consecutive 3
 - (iv) contains no three consecutive 5

Then the value of $\frac{a_{15}}{a_{10}}$ is

10. Let $P(x)$ be a non-constant polynomial of the least degree with real co-efficients satisfying $|P(-3)| = |P(-2)| = |P(-1)| = |P(1)| = |P(2)| = |P(3)| = 12$. Then the value of $|P(4)|$ is

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QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has two sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	One or more correct option(s)	8	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	-2 In all other cases	32
I(ii)	Paragraph Based (One or more correct option(s))	6	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	-1 In all other cases	24
II	Numerical Value Type (Up to second decimal place)	10	+3 If only the bubble corresponding to correct answer is darkened	—	0 If none of the bubbles is darkened	-1 In all other cases	30

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator